

Koopman-Based Online Identification With Sim2Real Transfer for Hydrodynamic Modeling of Turtle Inspired Robot

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Abstract—This letter addresses critical challenges in the dynamic modeling of biomimetic underwater robots, including heavy reliance on prior knowledge and difficulties in tracking time-varying models. To tackle these issues, we propose a control-oriented online identification framework based on the Koopman operator and successfully apply it to the hydrodynamics modeling of a sea turtle-inspired biomimetic robot. Initially, a prior model is obtained offline through simulation data based on Koopman theory. Subsequently, an online update algorithm integrated with Kalman filtering is developed to calibrate the model parameters in real time using a small amount of experimental data. Simulation and experimental results demonstrate that the proposed Koopman model outperforms baseline methods in long-term prediction accuracy and generalization capability. After online updates, the model achieves a 42.13% improvement in prediction accuracy over the baseline. In orientation tracking experiments under flow disturbances, the proposed framework enhances control accuracy by 63.4% compared to the baseline, while exhibiting smaller overshoot and stronger disturbance rejection capabilities. Overall, this study provides valuable insights for high precision dynamic modeling of biomimetic underwater robots.

Index Terms—Online system identification, bionic turtle robot, Koopman operator, Kalman filter.

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I. INTRODUCTION

BIO-INSPIRED underwater robots (BURs), leveraging their exceptional biological compatibility [1] and low environmental disturbance characteristics [2], have become indispensable tools in fields such as marine environmental monitoring [3] and archaeological exploration [4]. Compared to traditional propeller driven autonomous underwater vehicles (AUVs) and remotely operated vehicles (ROVs), BURs generate significantly lower noise [5], hydrodynamic wake [6], and sediment resuspension during locomotion [7]. For instance, Katzschmann et al. successfully utilized acoustic communication to control a robotic fish for close observation of marine life, providing an effective platform for studying organisms in their natural habitats [8].

However, the unique morphologies [9], numerous control degrees of freedom [10], and diverse bio-inspired propulsion mechanisms of these robots [11] pose significant challenges for precise dynamic modeling. To address this, researchers have proposed various methods [12]. Wang et al. established a three dimensional motion model for a robotic fish, including helical motion, based on Euler's equations of motion and grey box model hydrodynamic estimation [13]. Nonetheless, these methods typically rely on physical prior knowledge, including specific robot configurations, and determining model parameters often requires extensive time for meticulous design and experimental calibration. In previous work [14], although the single rigid body dynamic equations were employed to describe the motion characteristics of the amphibious turtle robot, the model still struggles to adequately capture the dynamic features arising from the three dimensional oscillation of the flippers.

Fortunately, data driven approaches, such as deep learning [15], sparse identification of nonlinear dynamics (SINDy) [16], and the Koopman operator [17], offer promising solutions for identifying nonlinear system behaviors solely from motion data. Deep learning involves solving nonlinear optimization problems and may suffer from convergence issues [15], while SINDy requires pre-constructing a dictionary of candidate functions [16], which can be restrictive for BURs with limited prior knowledge. In contrast, the Koopman operator framework globally linearizes nonlinear systems by lifting them into an infinite dimensional linear space while preserving control information. Koopman based system identification has been successfully applied to various systems, including spherical robots [18], soft continuum manipulators [19], and wheeled robots [20]. Notably, Pan et al. proposed a Bayesian Koopman framework for automatic

controller generation, effectively reducing the manual burden of constructing lifting functions, as validated on robotic fish platforms [21].

Despite reducing reliance on physical prior knowledge, Koopman-based methods still require substantial datasets for training [22]. Acquiring sufficient stable motion data for BURs is often challenging due to experimental constraints. A feasible strategy is to use dynamic simulators to generate diverse motion data for prior model identification, which is then updated and corrected with a small amount of real-world experimental data, enabling a Simulation-to-Reality (Sim2Real) transfer. This strategy has been widely adopted in embodied AI, including legged and humanoid robotics [23]. For example, the fine grained medium model proposed by Choi et al. enabled a reinforcement learning agent to achieve good adaptation to unfamiliar environments, overcoming limitations of the training data distribution [24].

Furthermore, BURs face the challenge of time varying dynamics in practical applications: factors such as varying payloads [18], mechanical wear of components [13], and coupling among multiple controllers [12] can lead to drift in dynamic parameters. Therefore, developing robust identification algorithms capable of online tracking of model variations is crucial. For instance, addressing the online identification of soft environmental dynamic models, Schindeler et al. proposed a two stage polynomial linearized Hunt-Crossley model identification method, achieving high precision online identification of nonlinear contact dynamics via a look up table recursive least squares estimator [25].

In summary, this study first identifies a Koopman model for the bio-inspired turtle robot based on simulation data. Subsequently, online updating of the model parameters is achieved using experimental data and a Kalman filter. The validity of the proposed framework is further validated through figure-eight path tracking simulations and directional tracking experiments. The main contributions are as follows:

- A Koopman operator modeling approach tailored for control tasks is proposed. The method significantly outperforms baseline methods in long-term prediction accuracy and generalization capability across different reference signals and control tasks. In simulations, the control accuracy is improved by 59.6% compared to baseline methods.
- An online update mechanism incorporating Kalman filtering is designed and implemented. This enhances the model's ability to track time-varying dynamics. On experimental datasets, the prediction accuracy is improved by 42.13% compared to baseline methods.
- Experimental validation of directional tracking is conducted in a pool with flow disturbances. The proposed framework achieves a 63.4% improvement in control accuracy over baseline methods, along with smaller overshoot and stronger disturbance rejection capability.

The letter is organized as follows: Section II introduces the preliminaries. Section III elaborates on the system identification algorithm based on Koopman theory. Section IV outlines the process of data driven identification and subsequent validation. Section V details the experimental implementation of an online Kalman filter based update mechanism for the Koopman model. Finally, Section VI presents the conclusions.

II. PRELIMINARIES

This section presents the design and mathematical modeling of the turtle inspired robot.

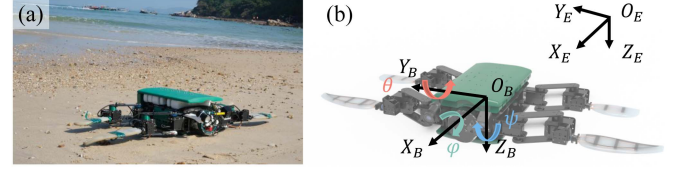


Fig. 1. Prototype and coordinate system. (a) The prototype of bio-inspired turtle robot. (b) An established definition of coordinate systems for the amphibious robotic platform, specifying the body fixed frame and the global inertial reference frame.

TABLE I
TECHNICAL SPECIFICATIONS OF THE BIO-INSPIRED TURTLE ROBOT

Item	Value & Description
Dimensions	$0.43 \times 1.23 \times 0.17 \text{ m}^3$
Weight	10.48 kg
Actuation Mechanism	3-DOF flippers
Sensing Modules	Camera, Pressure Sensors, Power Meter
Computing Unit	Raspberry Pi 4B (4 GB)
IMU	MPU-9250, 100 Hz sampling frequency
Servo Motors	GXYipin servo with 130 kg-cm torque
Endurance time	2 h
UWB	DW1000, accuracy error $\leq 100\text{mm}$

A. Design of the Turtle Inspired Robot

Turtles in nature, with their flattened body structure and powerful flippers, exhibit both excellent locomotor stability and efficient propulsion capabilities. Based on these biological features, we have designed a turtle inspired robot, whose prototype is shown in Fig. 1(a), with key structural parameters listed in Table I. The robot primarily consists of a central hull and a pair of symmetrically arranged three degree of freedom bionic flippers.

B. Mathematical Modeling

Fig. 1(b) illustrates the coordinate system for the turtle-inspired robot. The dynamic equation of the turtle inspired robot can be expressed as follows:

$$M(q)\dot{\mu} + C(\mu)\mu + D(\mu)\mu + g(q) = J^T F + S_a \tau \quad (1)$$

where q denotes the generalized coordinate and μ denotes the generalized velocities. The matrices $M(q)$, $C(\mu)$, and $D(\mu)$ represent the system inertia matrix, coriolis and centrifugal matrix, and hydrodynamic damping matrix, respectively. The vectors $g(q)$, F , and τ denote the gravitational and buoyancy forces/moments, external forces/moments, and generalized torque input, respectively. J represents Jacobian matrix toward applying points of F . The selective matrix S_a denotes the operations performed on τ .

Considering that the robot exhibits bilateral symmetry in both the longitudinal lateral (x - y) and longitudinal vertical (x - z) planes, the coupled motions among surge, sway, and yaw can be neglected. During locomotion, the turtle utilizes its forelimbs as the primary propulsion mechanism while the hindlimbs serve for directional control. Based on this biomimetic principle and engineering simplification objectives, and referring to recent study [26], we determine that only surge and yaw require active control in planar motion. Consequently, (1) can be simplified to a horizontal plane parametric hydrodynamic model that incorporates added mass, as well as linear and nonlinear damping

effects:

$$\begin{aligned}\dot{x} &= u \cos \psi - v \sin \psi \\ \dot{y} &= u \sin \psi + v \cos \psi \\ \dot{\psi} &= r \\ \dot{u} &= \frac{1}{m - X_{\dot{u}}} [X_u u + X_{u|u}|u| + \tau_X] \\ \dot{v} &= \frac{1}{m - Y_{\dot{v}}} [-(m - X_{\dot{u}})ur + Y_v v + Y_{v|v}|v|] \\ \dot{r} &= \frac{1}{I_{zz} - N_{\dot{r}}} [(X_{\dot{u}} - Y_{\dot{v}})uv + N_r r + N_{r|r}|r| + \tau_N]\end{aligned}\quad (2)$$

Here, x, y and ψ refer to the spatial configuration, while u, v and r denote the velocity components in the body-attached frame. $X_{\dot{u}}, Y_{\dot{v}}$, and $N_{\dot{r}}$ quantify added mass effects, while X_u, Y_v , and N_r correspond to linear damping coefficients. Quadratic damping terms $X_{u|u}, Y_{v|v}$, and $N_{r|r}$ capture nonlinear hydrodynamic resistance. The control inputs τ_X and τ_N govern surge propulsion and yaw torque respectively, with m representing total mass and I_{zz} the vertical axis moment of inertia. All hydrodynamic coefficients were obtained through CFD simulations, with the numerical values and detailed procedures provided in Reference [14].

C. Model Predictive Control

At each sampling instant t , the Model Predictive Control (MPC) solves the following finite-horizon optimal control problem over a prediction horizon of length N :

$$\begin{aligned}&\underset{\tau_{t|t}, \dots, \tau_{t+N-1|t}}{\text{minimize}} && \sum_{k=0}^{N-1} \ell(\chi_{t+k|t}, \tau_{t+k|t}), \\ &\text{subject to} && \chi_{t+k+1|t} = \chi_{t+k|t} + \Delta t \cdot f(\chi_{t+k|t}, \tau_{t+k|t}), \\ & && k = 0, \dots, N-1, \\ & && \chi_{t|t} = \chi(t), \\ & && \tau_{t+k|t} \in \chi, \quad k = 0, \dots, N-1, \\ & && \chi_{t+k|t} \in X, \quad k = 0, \dots, N-1,\end{aligned}\quad (3)$$

where $f(\chi, \tau)$ denotes continuous-time dynamics, $\tau \in \mathbb{R}^2$ and $\chi \in \mathbb{R}^6$ are control inputs and states respectively, and $\ell(\chi, \tau)$ is defined as:

$$\begin{aligned}\ell(\chi, \tau) &= Q_\psi (\psi - \psi_{ref})^2 + \tau^T R \tau \\ &\quad + (\tau - \tau_{prev})^T R_\Delta (\tau - \tau_{prev})\end{aligned}\quad (4)$$

with the following weighting matrices and parameters: $Q_\psi = 100$, $R = \text{diag}(1, 1)$ and $R_\Delta = \text{diag}(5, 1)$. After solving the optimization, only the first control input $\tau_{t|t}$ is applied to the system. The computational optimization process is accelerated by CasADi [27].

III. SYSTEM IDENTIFICATION

This section presents the system identification method based on Koopman theory and its online implementation via a Kalman filter.

A. The Koopman Operator

The Koopman operator advances measurement functions of the state with the flow of the dynamics. The Koopman operator $\mathcal{K}_t$ is an infinite dimensional linear operator that acts on eigenfunction ξ as:

$$\mathcal{K}_t \xi = \xi \circ F_t \quad (5)$$

where $\circ$ is the composition operator, t refers to the time step and F_t denotes a continuously mapping function.

With the Extended Dynamic Mode Decomposition (EDMD) method [9], the necessary observables are first selected to construct a finite dimensional space, followed by the extraction of eigenvalues from the data, thereby obtaining a finite dimensional approximate linear model. It should be noted that this model is not unique, and its specific form is entirely contingent upon the choice of observables.

Generally, the selection of observables primarily comprises three components: system states, inputs, and their associated functions. In multi-step prediction problems, to eschew introducing unnecessary complexity, the coupling functions between system states and inputs are typically not considered as independent observables. Based on this rationale, the evolutionary process of the system can be formulated as:

$$f[k+1] = \Theta^T \cdot f[k] \quad (6)$$

where $f[k]$ is the observables vector and Θ^T denotes the coefficient matrix that needs to be identified from data

The central goal of system identification is to minimize the fitting error, which can be mathematically formulated as minimizing the L^2 norm of the fitting error over all data.

$$\min_{\Theta} \sum_{k=1}^K \|\Theta^T \cdot f[k] - f[k+1]\|_2^2 \quad (7)$$

A linear koopman approximate model of the nonlinear system can be established by solving this least squares problem through methods such as singular value decomposition (SVD).

Path following is a common task for mobile robots, where the Line of Sight (LOS) method is widely applied to underactuated systems due to its ability to generate desired heading angles via waypoints, thereby reducing control complexity [4]. The heading degree of freedom is chosen as the control entry point, mainly based on the following two reasons: first, within the LOS framework, path following can be achieved solely through control of the heading degree of freedom; second, the key state variables corresponding to the heading degree of freedom—including the heading angle ψ and turning rate r —can be directly obtained from a single IMU sensor.

Based on the above considerations, this letter focuses on the heading degree of freedom for the control system design. To construct the system state observation framework, the state vector, control input, and reference signal at the current time step are defined as $\chi[k] = (\chi_x[k], \chi_y[k], \chi_\psi[k], \chi_u[k], \chi_v[k], \chi_r[k])^T \in \mathbb{R}^6$, $\tau_N[k] \in \mathbb{R}$, and $\zeta[k] \in \mathbb{R}$, respectively. Here, $\chi_x[k], \chi_y[k]$ denote the planar positions, $\chi_\psi[k]$ is the heading angle, and $\chi_u[k], \chi_v[k], \chi_r[k]$ represent the surge, sway, and yaw velocities. Subsequently, the lifted observable function vector $f[k]$ is constructed as follows:

$$f[k] = \begin{bmatrix} \chi_x[k], \chi_y[k], \chi_\psi[k], \chi_u[k], \chi_v[k], \chi_r[k], \\ \varepsilon[k], \chi_r[k]^2, \tau_N[k], \tau_N[k]^2 \end{bmatrix} \in \mathbb{R}^{10} \quad (8)$$

where $\varepsilon[k] = \zeta[k] - \chi_\psi[k]$ is the tracking error. Compared to generic basis functions such as radial basis functions (RBFs) or monomials, the proposed lifting functions are specifically tailored to the system's control objectives and excitation characteristics. This design enhances the Koopman model's predictive accuracy and interpretability without introducing unnecessary complexity or overfitting risks.

B. Online Identification Algorithm Based on the Kalman Filter

The coefficient matrix Θ in (6) varies over time in time varying systems. Conventional state estimation methods typically assume constant model parameters and employ the Kalman filter for state estimation. Given the assumption that the system state $\chi[k+1]$ acquired by sensors is sufficiently accurate, the Kalman filter can achieve optimal estimation of the parameter matrix Θ under noisy conditions based on observation data, namely the function $f[k+1]$ constructed from $\chi[k+1]$, thereby accomplishing online parameter identification. The design objective of the Kalman filter is to minimize the square of the posterior estimation error, which aligns with the optimization goal of online identification. Following the approach in reference [18], a separate Kalman filter is constructed for each equation to enable parallel online identification. The prediction model and observation function are as follows:

$$\text{Prediction: } \vartheta_i[k+1] = \vartheta_i[k] \quad (9)$$

$$\text{Observation: } f_i[k+1] = \vartheta_i^T[k+1]\mathbf{f}[k] \quad (10)$$

where ϑ_i is coefficient vector in Θ , $i = 1, \dots, N$, and N is the variable about system states. Generally speaking, Θ changes slowly and can be considered to be constant in a timestep. Therefore, the online system identification based on the Kalman filter can be finished as:

$$\vartheta_i[k+1|k] = \vartheta_i[k|k] \quad (9a)$$

$$P_i[k+1|k] = P_i[k|k] \quad (9b)$$

$$K_i[k+1] = \frac{P_i[k+1|k]\mathbf{f}[k+1]}{\mathbf{f}^T[k+1]P_i[k+1|k]\mathbf{f}[k+1] + R_i[k]} \quad (9c)$$

$$e_i[k+1] = f_i[k+1] - \vartheta_i^T[k+1|k]\mathbf{f}[k] \quad (9d)$$

$$\vartheta_i[k+1|k+1] = \vartheta_i[k+1|k] + K_i[k+1]e_i[k+1] \quad (9e)$$

$$P_i[k+1|k+1] = (I - K_i[k+1]\mathbf{f}^T[k+1])P_i[k|k] \quad (9f)$$

where P_i is the covariance matrix, K_i denotes the Kalman gain vector, R_i refers the variance, and e_i indicates the prediction error. To cope with the time-varying nature of noise statistics in real-world environments, an adaptive strategy is employed. The process noise covariance Q and measurement noise covariance R are initially tuned using offline data; during online operation, R is recursively estimated based on the innovation covariance, while Q is dynamically adjusted according to the trend of prediction errors, thereby maintaining optimal estimation performance. The complete update procedure of the Koopman model parameter Θ is detailed in the pseudocode provided in Algorithm 1.

IV. OFFLINE KOOPMAN MODEL IDENTIFICATION

This section outlines the data driven identification and subsequent validation of a dynamic model for a bio-inspired turtle robot, leveraging Koopman operator theory.

A. Data Collection and System Identification

Data driven methods for dynamic model identification place high demands on data quality. To meet this requirement, this study utilizes the dynamic simulator Webots to acquire high fidelity motion data of a bio-inspired turtle robot. To generate a

Algorithm 1: Online Koopman Model Identification via Kalman Filter.

Require: Offline identified parameters $\vartheta_i[0]$, experimental data from MPC tracking, forgetting factor λ , scaling factor α , adjustment period L , bounds $[Q_{\min}, Q_{\max}]$

Ensure: Updated Koopman model parameter matrix Θ

- 1: Initialize $P_i[0] \leftarrow I$, $Q \leftarrow Q_0$, $R \leftarrow R_0$, $\mathbf{f}[0]$, $S_{\text{innov}}[0] \leftarrow \mathbf{0}$
- 2: $K_T \leftarrow$ total time steps, $N \leftarrow 10$
- 3: **for** $k = 0$ to $K_T - 1$ **do**
- 4: Read new measurements $\chi[k+1]$, $\zeta[k+1]$, $\tau[k+1]$
- 5: Compute $\varepsilon[k+1] \leftarrow \zeta[k+1] - \chi_\psi[k+1]$
- 6: Construct $\mathbf{f}[k+1] \leftarrow (8)$
- 7: Innovation: $\nu[k+1] \leftarrow \mathbf{y}[k+1] - \hat{\mathbf{y}}$
- 8: Update innovation covariance: $S_{\text{innov}}[k+1] \leftarrow \lambda S_{\text{innov}}[k] + (1 - \lambda)\nu[k+1]\nu[k+1]^T$
- 9: Adapt measurement noise:
 $R \leftarrow \alpha \text{diag}(S_{\text{innov}}[k+1])$ (if $k < 10$, keep $R = R_0$)
- 10: **for** $i = 1$ to N **do**
- 11: Kalman filter update for ϑ_i using 9
- 12: Update $\Theta[:, i]$
- 13: **end for**
- 14: Periodic Q adaptation: if $\text{mod}(k+1, L) = 0$, compute recent error trend Δe ; if $\Delta e > \delta$, $Q \leftarrow \min(\beta Q, Q_{\max})$; if $\Delta e < -\delta$, $Q \leftarrow \max(\gamma Q, Q_{\min})$.
- 15: $\mathbf{f}[k] \leftarrow \mathbf{f}[k+1]$
- 16: **for** $i = 1$ to N **do**
- 17: $\vartheta_i[k|k] \leftarrow \vartheta_i[k+1|k+1]$
- 18: $P_i[k|k] \leftarrow P_i[k+1|k+1]$
- 19: **end for**
- 20: **end for**

dataset that sufficiently excites the system's dynamic characteristics, path following task and heading angle tracking task are designed. The specific tasks are illustrated in Fig. 2(a).

Using this dataset, an offline Koopman model is identified by solving the least squares problem described in (7) with the lifting functions defined in (8). To validate the open-loop prediction accuracy of the model, the representative horizontally aligned figure "8" path is selected as the validation set. As a benchmark for comparison, the parameters of the dynamic model given in (2) in Section III are also directly identified using the least squares method based on the same dataset, following the strategy consistent with Reference [28]. Finally, a comparative analysis of the open loop heading angle predictions from the parametric CFD model, the least squares identified model, and the Koopman model is presented in Fig. 3. The reference signal serves as an external input in the Koopman model, whereas the baseline methods are driven solely by the initial state and control inputs.

Furthermore, to comprehensively evaluate the generalization capability of the model, predictions were made for the heading angle variations across four prediction sets, including the horizontally aligned figure "8" path, and the results are presented in Fig. 3(c). The comparison demonstrates that the proposed Koopman model achieves significantly higher prediction accuracy than the other two baseline models. For instance, in predicting the heading angle for a heartshaped trajectory, the average

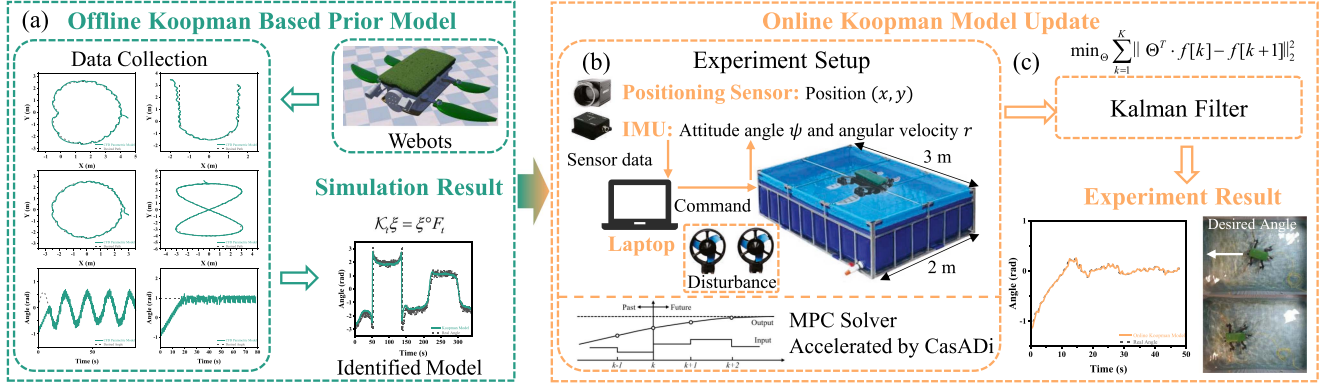


Fig. 2. Koopman operator based Sim2Real transfer framework. (a) Offline Koopman model identification from simulation data. (b) Experimental setup for real-world data acquisition. Location data can be acquired through various sensors, such as GPS for surface applications or acoustic equipment for underwater positioning. In this letter, a camera was utilized in convenience of experimental setup. (c) Online Koopman model updating via experimental data and a Kalman filter.

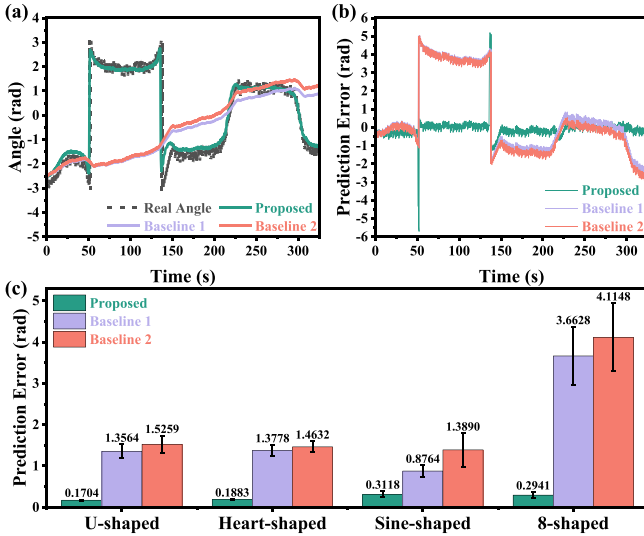


Fig. 3. Analysis of the offline model accuracy. (a) Comparison of heading angle predictions among different identification models in figure-eight path following. (b) Comparison of heading angle prediction errors for the corresponding models. (c) Comparison of heading angle prediction errors for different data type, with error bars calculated from five simulation trials. Proposed: Koopman Model. Baseline 1: CFD Parametric Model. Baseline 2: Least Square Identified Model.

prediction error of the Koopman model is only 0.1883 rad, which is substantially lower than the 1.3778 rad of the CFD based parametric model and the 1.4632 rad of the least squares identified model.

The results indicate that only the Koopman model can accurately identify the abrupt changes in the robot's yaw angle during motion and satisfactorily track its variation trend. In contrast, the predictions from the parametric CFD based model and the least squares identified model exhibit smoother variations, suggesting their difficulty in capturing the dynamic transients of the system's motion.

B. Validation of the Generalization Capability

To effectively validate that the proposed method encodes the essential dynamic behavior of the turtle-inspired robot and

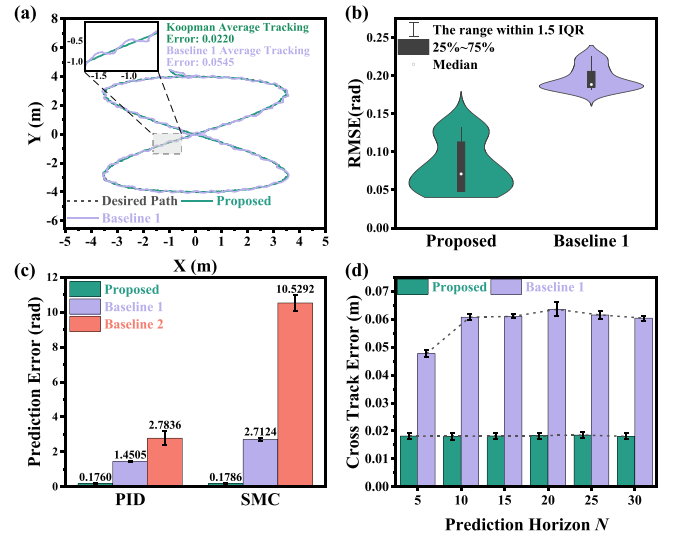


Fig. 4. Validation of the generalization capability. (a) Trajectory comparison between the proposed method and baseline method in figure-eight path following. (b) Violin plots of MPC tracking error comparison between the proposed method and baseline method under random reference signals. (c) Prediction comparison of the proposed method and baseline method for trajectories generated by different controllers, with error bars calculated from five simulation trials. (d) Tracking error comparison between the proposed method and baseline method under different prediction horizons, with error bars calculated from five simulation trials. Proposed: Koopman Model. Baseline 1: CFD Parametric Model. Baseline 2: Least Square Identified Model.

generalizes well across varying reference inputs and control laws, simulation analysis is conducted as shown in Fig. 4. Under an identical control framework and parameter set, the tracking trajectory of the proposed method is smoother; quantitative analysis reveals an average lateral tracking error of 0.0220 m, which is significantly lower than the 0.0545 m error of the parametric CFD model, achieving an improvement of accuracy by 59.6%.

Following the principle of Pseudo-Random Binary Sequence (PRBS), we generated diverse reference signals and used the same MPC controller to track them. The results demonstrate a narrower interquartile range and a more concentrated error distribution, indicating that Koopman model captures intrinsic

dynamic features rather than memorizing reference-specific behaviors, as shown in the violin plot of Fig. 4(b). To further assess generalization across different controllers, we employed the learned Koopman model to predict robot motion under two unseen control strategies: PID and Sliding Mode Control (SMC). As illustrated in Fig. 4(c), the proposed method maintained superior prediction performance in both cases, confirming that the model is not overfitted to a particular controller, but instead captures the underlying open-loop dynamics of the robotic system.

To analyze the impact of the model's long-horizon prediction accuracy on the controller, this letter examines the path-tracking precision of the proposed method and the baseline method under different prediction horizons, with results presented in Fig. 4(d). The average cross tracking error of the proposed method remains consistently stable at approximately 0.018 m, indicating that the error does not significantly accumulate with the extension of the prediction horizon. In contrast, for the baseline method, the tracking error rises markedly as N increases from 5 to 10, directly reflecting that model mismatch exacerbates the error as the prediction horizon expands, thereby reducing the control performance of the MPC.

V. ONLINE KOOPMAN MODEL UPDATE

This section details the experimental implementation of an online Kalman filter (KF) based update mechanism for the Koopman model.

A. Experiment Setup and Online Model Update

Simulation data often fails to capture the full complexity of the real world, particularly the influence of environmental disturbances such as waves and flows on robot dynamics. To address this issue and achieve rapid acquisition of a high precision model, this study introduces an online update mechanism built upon a prior model established from simulation data.

The experimental setup is shown in Fig. 2(b). The robot's pose is obtained through multi sensor fusion: an overhead camera fixed above the water pool provides positional coordinates, from which the velocity is derived via differentiation; an onboard IMU directly measures the yaw angle and yaw rate. To evaluate the real time predictive capability of the online updated model, we employ a one step ahead prediction to test the model performance. The primary objective is to determine whether the final model after online updating outperforms the original offline model. The Recursive Least Squares (RLS) algorithm is selected as a benchmark for comparison. This method achieves parameter updates recursively through a mathematical transformation. In contrast, the Kalman filter based update method adopted in this work essentially minimizes the posterior estimation error.

As outlined in Algorithm 1 and Fig. 2(c), the online parameter update is performed. The results are presented in Fig. 5(a). The model updated via the Kalman filter responds rapidly and closely tracks the system's dynamic changes. In comparison, the original offline model and the model updated using RLS exhibit sluggish tracking and inadequate response. As shown in the logarithmic scale tracking error plot in Fig. 5(b), the proposed Kalman filter based update method achieves a significantly smaller mean prediction error of 0.0328 *rad*, compared to 0.0566 *rad* for the RLS method and 0.0988 *rad* for the offline model. In this study, five sets of directional tracking data were collected. The average

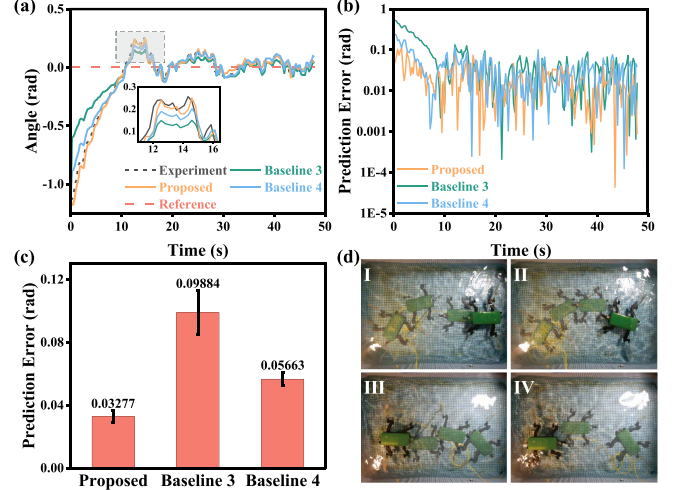


Fig. 5. Analysis of the online model accuracy. (a) Multi model comparison of one step predictions for heading tracking. (b) Comparison of yaw tracking error in logarithmic scale. (c) Online average prediction errors for various models on five experimental datasets, with error bars indicating the standard error. (d) Selected experimental snapshots. Proposed: Koopman Model + KF. Baseline 3: Offline Koopman Model. Baseline 4: Koopman Model + RLS.

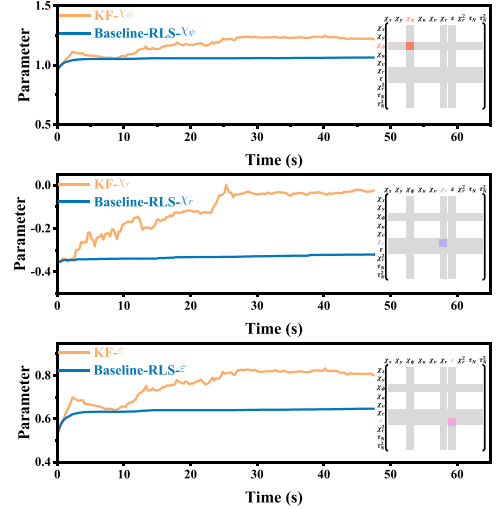


Fig. 6. Comparison of online updates for Key Koopman model parameters.

prediction errors are presented in Fig. 5(c). Compared with the RLS method, the proposed framework improves the prediction accuracy by 42.13%. Selected experimental snapshots are shown in Fig. 5(d).

Furthermore, Fig. 6 illustrates the temporal variation of key model parameters (ψ and r) during the yaw angle tracking task. The figure shows that the Kalman filter method applies more substantial updates to the parameters, and the parameter values converge to stability more quickly. This further validates the effectiveness of the Kalman filter method in rapidly converging to and stably estimating model parameters in dynamic environments.

B. Verification With MPC in Experiment

To evaluate model performance under disturbed conditions, this study conducted experiments in a water tank by arranging

TABLE II
ABLATION STUDY OF THE PROPOSED FRAMEWORK

Architecture	MAE (rad)	RMSE (rad)	MaxAE (rad)	STD (rad)	90% Percentile Error (rad)
Full Proposed	0.0328	0.0438	0.1346	0.0292	0.0723
Least Square + Sim2Real + KF	0.0581	0.0675	0.1666	0.0344	0.0992
Koopman + RLS	0.0566	0.0756	0.2487	0.0502	0.1229
Offline Koopman	0.0988	0.1402	0.4534	0.0986	0.2408

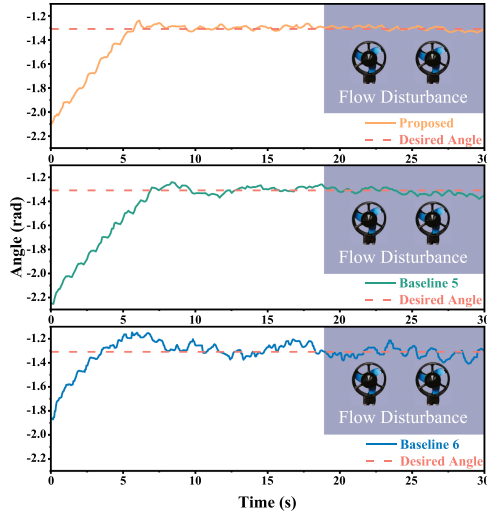


Fig. 7. Variations in the heading angle of the turtle inspired robot when performing directional tracking tasks. Proposed: MPC with online updated koopman model; Baseline 5: MPC with offline koopman model; Baseline 6: MPC with CFD parametric model.

propellers to generate controllable flow disturbances, thereby simulating the uncertain dynamics encountered in real-world operating conditions. To ensure comparability among different dynamic models, comparative experiments on orientation tracking based on MPC were carried out under consistent controller parameter settings.

The experimental results are shown in Fig. 7. Compared with the CFD parametric model, both the offline Koopman model and the online Koopman model exhibited lower overshoot and demonstrated better orientation maintenance performance under flow disturbances. Among them, the proposed online Koopman model further improved tracking stability, indicating its effectiveness in capturing unmodeled dynamics not represented in simulations. During the steady state tracking phase, the proposed method achieved a root mean square error (RMSE) of 0.0164 rad , which was lower than the 0.0294 rad of the offline Koopman model and the 0.0448 rad of the CFD parametric model. Compared with the baseline methods, the tracking error of the proposed approach was reduced by 63.4%, validating the effectiveness of the online update mechanism in enhancing model prediction accuracy and disturbance rejection capability. Fig. 8 displays a sequence of the robot's motion postures during the experiment, visually illustrating its movement and adjustments within the confined space.

C. Ablation Study

The ablation study results presented in Table II quantitatively demonstrate the contribution of each component within the proposed framework. The data in Table II are averaged over five

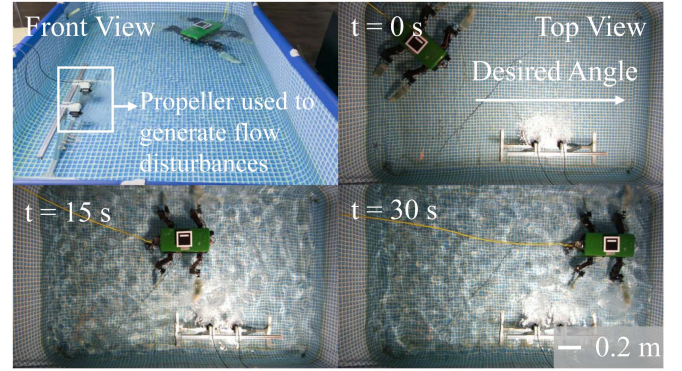


Fig. 8. Experimental snapshots of the turtle inspired robot during a directional tracking task.

experimental runs, with the selected metrics being mean absolute error (MAE), RMSE, maximum absolute error (MaxAE), standard deviation of error (STD), and the 90% percentile error. First, the necessity of online adaptation is evidenced by the fact that the Koopman model updated with RLS significantly outperforms the offline Koopman model, confirming that simulation data alone cannot fully capture real-world dynamics. Second, the superiority of the Kalman filter as an online update mechanism is validated by the full proposed method, which achieves the lowest errors across all metrics (MAE = 0.0328, RMSE = 0.0438), surpassing the Koopman+RLS variant. Finally, the importance of Koopman lifting is highlighted by comparing the full proposed method with the least squares-identified model: despite both employing Sim2Real pre-training and Kalman filtering, the Koopman-based architecture yields substantially lower prediction errors (e.g., MAE 0.0328 vs. 0.0581). These comparisons collectively affirm that each module—Koopman lifting, Sim2Real pre-training, and Kalman-based online update—indispensably contributes to the overall performance of the framework.

VI. CONCLUSION

In this letter, we propose an online system identification method based on the Koopman operator and integrate it within a Sim2Real transfer framework, which is successfully applied to the hydrodynamic modeling of a turtle inspired robotic system. The online algorithm, designed based on Kalman filter, enables rapid and accurate online updates of the offline trained prior model using limited real world experimental data, effectively capturing the time varying characteristics of the system in dynamic environments. Experimental results demonstrate that the proposed Koopman model outperforms traditional parametric CFD model and least squares identified methods in both yaw motion prediction and control tasks, while maintaining strong

robustness and tracking accuracy even under disturbances generated by propellers. Moreover, by adopting the Sim2Real transfer strategy, we significantly reduce the reliance on extensive real motion data and enhance the applicability of data driven modeling in scenarios with limited experimental conditions.

Future work will focus on the translation-invariant formulation of the Koopman model that relies only on velocities and error states.

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